

NoteNova Autonomous AI Ecosystem — Master Architecture

Multi-Agent Infrastructure · Meta Graph API · Groq Llama-3.3 · Razorpay · Xenova RAG · Web Speech

EXECUTIVE OVERVIEW

📄 Overview of the 5 Autonomous Core Systems

This technical document details the end-to-end execution flows, routes, schemas, and AI models powering the 5 specialized autonomous agents in the NoteNova ecosystem.

1. AI REVENUE RECOVERY AGENT

Autonomous 5-signal financial engine. Captures failed checkouts & OTP timeouts, runs Groq Llama-3.3 diagnosis, evaluates policy gates ($\leq ₹5,000$ auto vs $> ₹5,000$ HITL), mints 24h vouchers, and recovers revenue via WhatsApp & in-app alerts.

`/api/agent/revenue-recovery/*`

2. INSTAGRAM GROWTH AGENT

1-Click direct Meta Graph API publisher for 8K Reels. Auto-schedules daily morning drafts at 06:00 AM IST, polls video containers until FINISHED status, publishes live, and purges Cloudinary storage automatically.

`/api/instagram-agent/*`

3. AI CLIENT ACQUISITION AGENT

B2B client project pipeline. Ingests leads, analyzes technical requirements with AI, drafts milestone proposals, dispatches WhatsApp/Email outreach, generates public escrow payment links (`/pay/:code`), and builds codebases.

`/api/client-agent/*`

4. POCHI (ADMIN VOICE ASSISTANT)

3D glowing Siri/Orb executive hologram. Speech-to-Text & Speech-to-Text interaction equipped with direct database mutation tools to modify batch prices, review refunds, and inspect live revenue analytics.

`/api/admin/pochi/*`

5. AI ACADEMIC CHATBOT & STUDY COPILOT

Omniscient student doubt solver built with local Xenova/all-MiniLM-L6-v2 vector embeddings (RAG). Intercepts PDF summarization (`__PDF_SUMMARISE__`), detects in-chat Promise-to-Pay commitments, diagnoses live payment declines, and provides real-time course recommendations from live MongoDB catalog.

`/api/chatbot`

🔧 Technology & Model Stack

Primary Cognitive LLM: Groq Cloud SDK (llama-3.3-70b-versatile, qwen3.8-27b)

Payment Gateway: Razorpay Node SDK (HMAC-SHA256 Signed Webhooks)

Voice Engine: Web Speech API (webkitSpeechRecognition + SpeechSynthesis)

Fallback LLM: OpenRouter API (nvidia/nemotron-nano-9b-v2:free)

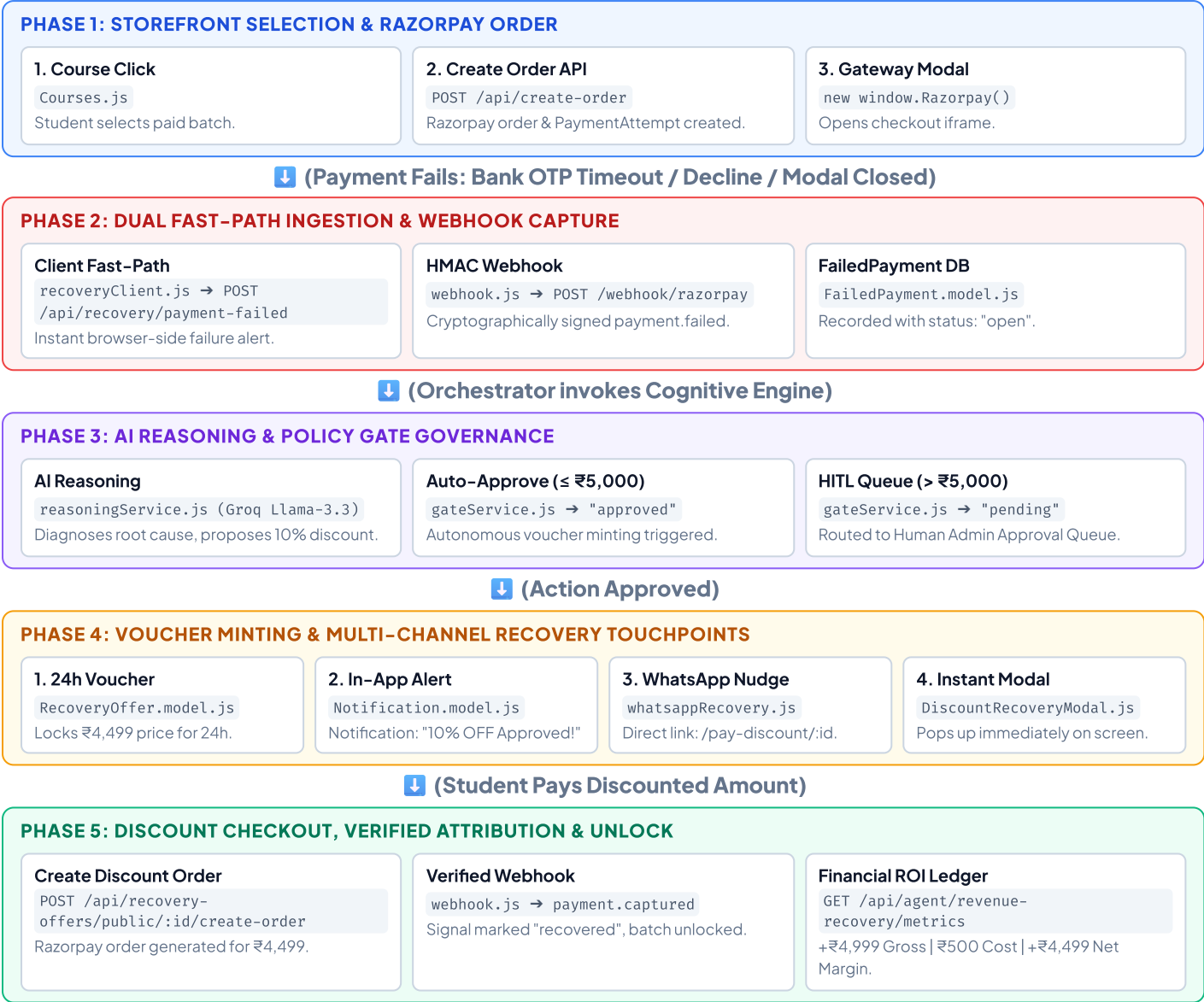
Vector Embeddings: @xenova/transformers (all-MiniLM-L6-v2)

Social Engine: Meta Graph API v21.0 (Instagram Business Container API)

1. AI Revenue Recovery Agent — Architecture & Flowchart

End-to-End Failure Detection, Cognitive Diagnosis, Policy Gating, Voucher Minting & ROI Attribution

REVENUE AGENT

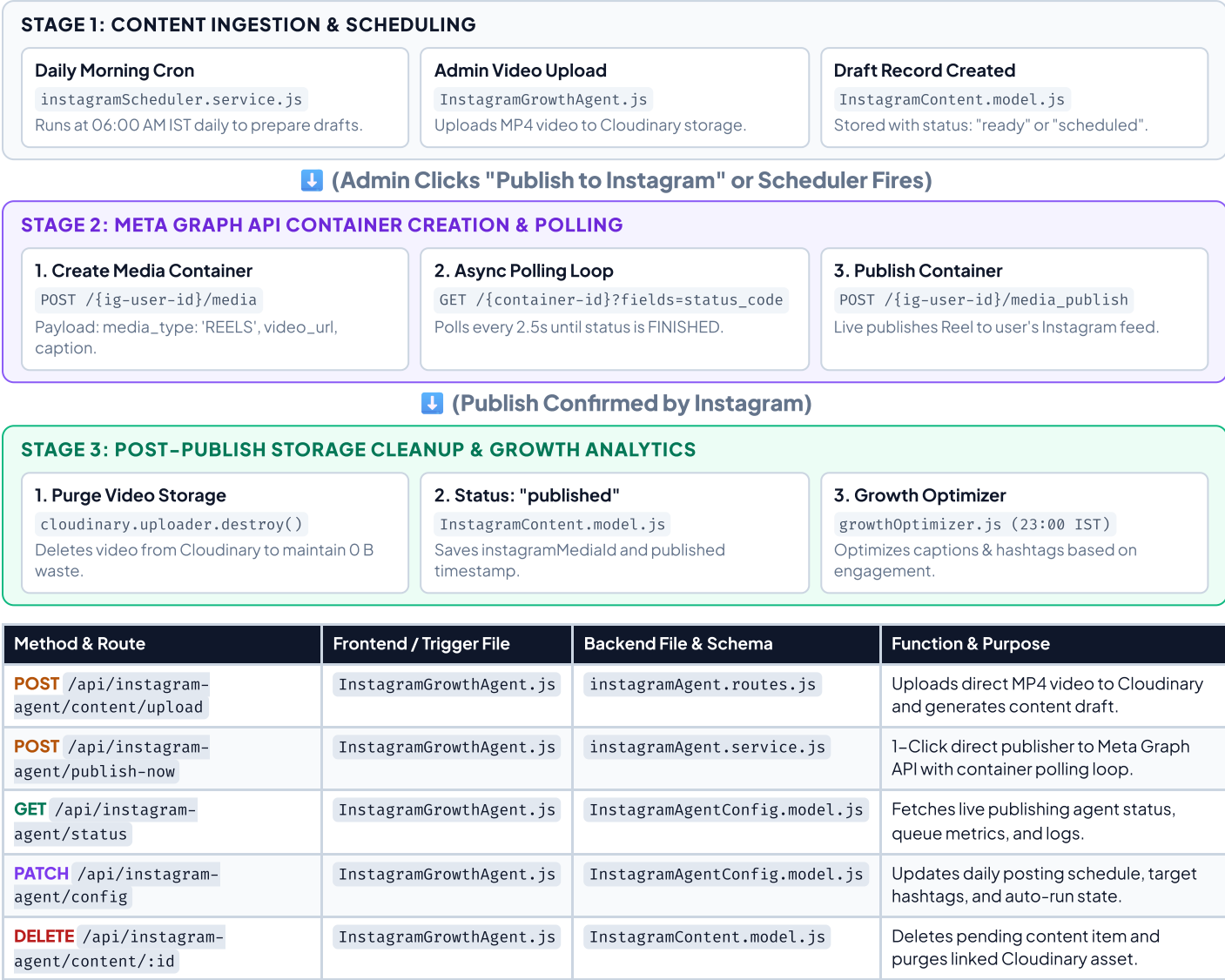


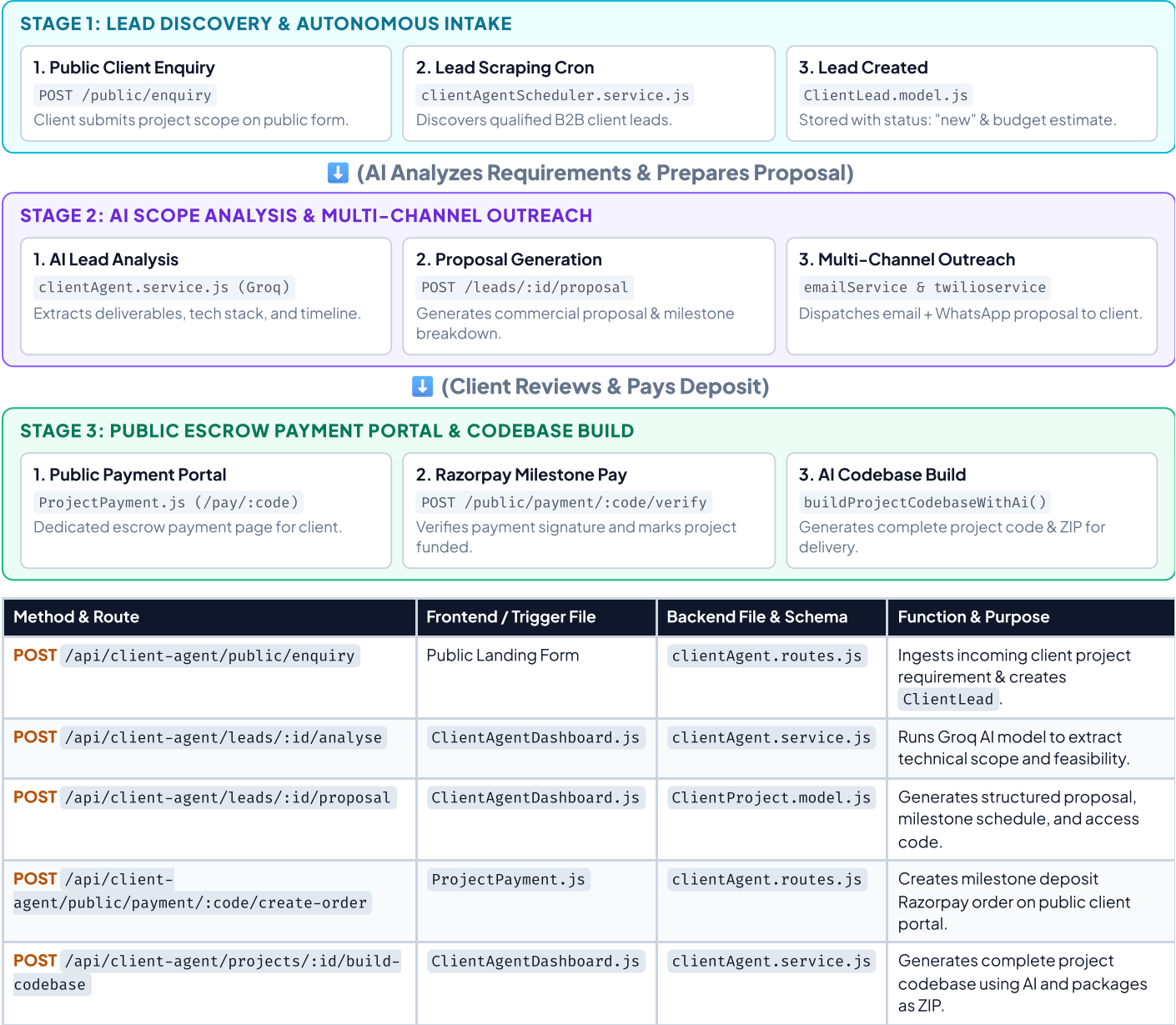
Method & Route	Frontend / Trigger File	Backend File & Schema	Function & Purpose
POST /api/create-order	Courses.js	back/index.js (Line 238)	Creates initial Razorpay order and logs PaymentAttempt .
POST /api/recovery/payment-failed	recoveryClient.js	recoveryClient.routes.js (Line 17)	Fast-path failure reporter; immediately evaluates gate and returns discount.
POST /api/agent/revenue-recovery/webhook/razorpay	Razorpay Webhook Cloud	webhook.js (Line 18)	HMAC-verified webhook listener for payment.failed & payment.captured .
POST /api/recovery-offers/public/:id/create-order	DiscountRecoveryModal.js	recoveryOffer.routes.js (Line 33)	Creates discounted Razorpay order for verified 24h recovery voucher.
GET /api/agent/revenue-recovery/metrics	RevenueRecoveryDashboard.js	revenue-recovery/routes.js (Line 176)	Calculates real-time Gross Revenue, Net Profit Margin, and Conversion %.

2. Instagram Growth Agent — Architecture & Flowchart

1-Click Direct Meta Graph API Publisher · Video Container Polling · Storage Cleanup

INSTAGRAM AGENT





 **SYSTEM 4: POCHI — ADMIN EXECUTIVE 3D VOICE AI**

1. Voice Input (STT)
PochiVoiceAssistant.js
Web Speech API captures voice in English/Hindi.

2. Pochi Query API
POST /api/admin/pochi/query
Sends voice query with full admin context.

3. Tool Execution
pochiAdminContext.js
Direct DB mutation: edits price, refunds, doubts.

4. Hologram Speech (TTS)
Pochi3DViewerModal.js
3D Orb visualizer pulses as voice replies.

 **SYSTEM 5: AI ACADEMIC CHATBOT & STUDY COPILOT (RAG)**

1. Message / PDF Upload
dash/src/features/pochi/chatbot
Student asks academic question or uploads PDF.

2. Intent Interceptors
studentContext.js
Fast-path: PDF Summary, P2P date, Payment status.

3. Vector Search (RAG)
Xenova/all-MiniLM-L6-v2
Cosine similarity on website_knowledge.md chunks.

4. Omniscient Context
Groq Llama-3.3-70b
Injects live course catalog, price in ₹, user history.

Method & Route	Frontend / Trigger File	Backend File & Schema	Function & Purpose
POST /api/admin/pochi/query	PochiVoiceAssistant.js	pochi.routes.js (Line 18)	Admin-only voice assistant endpoint; executes direct DB tools.
GET /api/admin/pochi/context	PochiVoiceAssistant.js	pochiAdminContext.js	Fetches live store snapshot (revenue, doubts, active students) for Pochi.
POST /api/chatbot	features/pochi/chatbot.js	modules/pochi/chatbot.js (Line 291)	RAG academic study copilot, PDF summarizer, and live course advisor.

 **Key Features of the Chatbot & Pochi**

 **Promise-to-Pay in Chat:** If a student says "main kal pay karunga", the chatbot registers the commitment in FailedPayment, pauses recovery alerts, and confirms their seat.

 **Pochi Database Tools:** Direct mutation of MongoDB records (Batch.price, RefundRequest, doubt responses) via voice.

 **PDF Summarization:** Parses uploaded notes using pdf-parse (up to 15,000 characters) and generates structured takeaways.

 **Omniscient Catalog Injection:** Chatbot checks real-time prices and never recommends courses the student already owns.